

Sable Arm Power Budget

Team Number:	303
Project Name:	Sable
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Version:	2

All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	9G RC Servo	SER0039	+0-5V	3	300	900	mA
	Stepper Motor	42BYG40B-18T42	+3.4V	1	1700	1700	mA
	Stepper Motor driver	TMC2209-LA-T	+3-5.25V	12	12	12	mA
	LED	LTST-M670TBKT	+0-3.3V	5	20	100	mA
	ESP32 Microcontroller	ESP32-S3-WROOM-1	+2.7-3.6V	1	355	355	mA
+12V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	Stepper Motor	42BYG40B-18T42	+3.4V	1	1700	1700	mA
	Subtotal					1700	mA
	Safety Margin					25%	
	Total Current Required on +12V Rail					2125	mA
c1. Regulator or Source Choice	+12V Battery		N/A	1	5000	5000	mA
Total Remaining Current Available on +12V Rail						2875	mA
+5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	9G RC Servo	SER0039	0-5V	3	300	900	mA
	Subtotal					900	mA
	Safety Margin					25%	
	Total Current Required on +5V Rail					1125	mA
c2. Regulator or Source Choice	+5V Buck Converter	LM2592HVS-5.0	+0-57V	1	2000	2000	mA
Total Remaining Current Available on +5V Rail						875	mA
+3.3V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	ESP32 Microcontroller	ESP32-S3-WROOM-1	+2.7-3.6V	1	355	355	mA
	Stepper Motor Driver	TMC2209-LA-T	+0-5V	1	25	25	mA
	LED	LTST-M670TBKT	+0-3.3V	5	20	100	mA
	Subtotal					480	mA
Safety Margin					25%		
Total Current Required on +3.3V Rail					600	mA	
c3. Regulator or Source Choice	+3.3V Buck Converter	LM2575-3.3WU-TR	+4-40V	1	1000	1000	mA
Total Remaining Current Available on +3.3V Rail						400	mA
External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output Voltage	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
Power Source 1 Selection	+12V Battery	N/A	N/A	+12V	5000	5000	mA
Power Rails Connected to External Power Source 1	+12V Power Rail	N/A	N/A	+12V	1662.5	1662.5	mA
	+5V Buck Converter	LM2592HVS-5.0	+0-57V	+5V	2000	2000	mA
	+3.3V Buck converter	LM2575-3.3WU-TR	+4-40V	+3.3V	1000	1000	mA
	Total Remaining Current Available on External Power Source 1					337.5	mA
External Power Source 2	Component Name	Part Number	SupplyVoltageRange	Output Voltage	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
Power Source 2 Selection	+9V 5A Wall Power Supply	0930	90-264 VAC	9V	3000	3000	mA
Power Rails Connected to External Power Source 2	+12V Power Rail	N/A	N/A	+12V	1662.5	1000	mA
	+5V Buck Converter	LM2592HVS-5.0	+0-57V	+5V	2000	900	mA
	+3.3V Buck converter	LM2575-3.3WU-TR	+4-40V	+3.3V	1000	500	mA
	Total Remaining Current Available on External Power Source 2					600	mA
Battery Life Calculation	Component Name	Part Number	SupplyVoltageRange	Capacity(mAh)		RequiredByRegulators	
	Battery	BGN5500-10EWP-A800EC	+12V	5500		2400	mA
	Battery Life					2.291666667	hours